

2026 IEEE SUSTAINABLE POWER AND ENERGY CONFERENCE

iSPEC 2026

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SPECIAL SESSION 2 (SS2) • AHGPC-TCR

Advanced High-Gain Power Converters: Topologies, Control, and Renewable Energy Integration

— Call for Papers —

Organized and Co-Chaired by:

- **Dr. Seshagiri Rao Vemparala** — Postdoctoral Researcher, Renewable Energy Lab, Prince Sultan University, Saudi Arabia
- **Dr. Anjana M. P.** — Application Engineer, Infinipoint Technologies Pvt. Ltd.
- **Dr. Kumaravel S.** — Professor, Dept. of Electrical Engineering, NIT Calicut

Session Overview:

High-gain power converters are key enablers for efficient, reliable, and compact power conversion in renewable energy and electrified systems. This session focuses on switched-inductor, switched-capacitor, voltage-multiplier, coupled-inductor, and hybrid converter structures that achieve high voltage gain with reduced component stress and losses, alongside advanced control, MPPT, and energy management for solar PV, wind, storage, DC microgrids, and electric mobility — connecting converter research with iSPEC 2026's carbon-neutrality theme.

Topics of Interest Include, but Are Not Limited to:

- Novel high-gain DC-DC converter topologies for renewable energy
- Switched-inductor, switched-capacitor, and voltage-lift converter techniques
- Coupled-inductor-based high-gain DC-DC converters
- Quadratic, cubic, and other high-order voltage-gain structures
- High-efficiency, soft-switching, resonant, and wide-bandgap converters
- Bidirectional high-gain converters for batteries and hybrid storage
- Model predictive, nonlinear, adaptive, and AI/ML-based converter control
- MPPT and voltage/current regulation for renewable energy systems
- Power converters for EVs, charging infrastructure, and V2G applications
- Reliability, thermal management, and fault tolerance of converters

Important Dates:

Paper Submission Opens	25 August 2026
Paper Submission Deadline	30 September 2026
Acceptance Notification	15 October 2026
Registration Due	1 November 2026
Conference Dates	4–6 December 2026

Format: Half-day technical special session, targeting ~8–10 peer-reviewed papers. Author guidelines and peer-review standards will follow the regular iSPEC 2026 paper submission process. Accepted and presented papers will be submitted for inclusion in IEEE Xplore, subject to IEEE Xplore's scope and quality requirements.

Submission Portal: <https://ispec2026.org/submission>

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